



Assessment of blurred vision

History

- **Does the patient genuinely mean BLURRED (loss of clarity or sharpness of vision)?** Other things sometimes described as blurriness include scotomas (visual field defects, although patients rarely aware of these!), diplopia, floaters and flashes (see our article on these) and metamorphosia (visual distortion).
- Sudden or gradual?
- How long has it been present for?
- One or both eyes? (unilateral, may present late in a disease process because other eye compensates).
- Does it improve with blinking? *In the vast majority, this means the diagnosis is dry eyes.*

Associated symptoms

- Flashes and floaters? New or persistent? (*means vitreous is detaching from retina*)
- Visual 'curtain' – sudden-onset sensation of a dark curtain falling across one eye)? (*retinal detachment ?TIA*)
- Visual distortion?
- Loss of central vision? (*means fovea is affected*)
- Pain (see article on red eye if painful red eye):
 - Is it painful to move the eye? How severe?
 - Sharp sensation as if foreign body present?
 - Any recent trauma?
- Headache? (*look for scalp tenderness, temporal pain, jaw claudication*)
- Redness?
- Discharge?

PMH

- AGE! Sudden-onset blurred vision in an 80-year-old is more likely to be GCA, whereas in a 30-year-old it is more likely to be MS.
- Systemic disease: diabetes, hypertension, autoimmune disease such as the seronegative arthropathies (can cause uveitis).

Ophthalmic history

- Occupation with high risk of foreign bodies in eye? (especially power tools and hammering!)
- Contact lens user?
- Recent surgery or intra-vitreous injections?
- Any history of amblyopia (lazy eye)?

Family history

- Glaucoma?
- Inherited eye disease, e.g. retinitis pigmentosa ?

Examination

- **Visual acuity** (in both eyes separately!); use a pinhole if glasses left at home!
- **External appearance of the eye:**
 - Eyelids (periorbital cellulitis?).
 - Orbital involvement? (proptosis, restriction of eye movement)
 - Conjunctiva:
 - Uniform redness more likely to suggest conjunctivitis.
 - Sectorial may be episcleritis.
 - Circum-corneal injection may suggest anterior uveitis.
- **Eye movements**
 - Check for full range of eye movements.
 - Painful movements and evidence of optic nerve dysfunction (blurriness, loss of colour vision, relative afferent pupillary defect, see below) suggest optic nerve pathology.
- **Double vision**
 - **Monocular diplopia** (double vision persisting with one eye closed): suggests a problem in the eye itself (*usually a cataract*).
 - **Binocular double vision** (double vision that goes when one eye is closed): suggests ocular malalignment. If acute, suggests cranial nerve palsy or intracranial lesion.
- **Pupil reactions**
 - **Size:** check similar size. If irregular, there are several causes including 3rd nerve palsy, uveitis (but this is usually painful) and Horner's syndrome. Remember: 3rd nerve, fixed pupil: likely compressive: surgical emergency. 3rd nerve palsy with accommodating pupil: likely ischaemic.
 - Relative afferent pupillary defects: very difficult to examine for in practice so leave for the specialist!
- **Fundus examination:** for red reflex (absence suggests gross pathology in that eye) and papilloedema.
- **Fluorescein staining** to assess corneal epithelium and look for ulcers, etc.; examine under a blue light!

Blurred vision

2. Diagnosis and management

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Diagnosis and management

Put simply, management is:

Sudden onset: usually urgent referral/discussion with ophthalmology (or via TIA/stroke pathway if indicated).

Gradual onset: usually routine ophthalmology referral (or optician for refractive errors). (Note that many patients with gradual loss will report this as sudden loss!)

RED EYES (painful and painless) are covered in a separate article, although some of the causes listed below also present as a red eye.

Sudden onset (immediate up to a few days)	Unilateral	Painful	- Trauma to the eye (do NOT palpate globe – may rupture!). - Corneal ulcer, including ulcer in contact lens user and ulcer from herpes simplex or zoster (see article in the online handbook). - Acute angle closure glaucoma. - Uveitis. - Orbital cellulitis or endophthalmitis (internal eye inflammation, usually from infection, most commonly after surgery/intra-vitreous injections; presents as painful red eye, often with nausea/vomiting and loss of vision).	Urgent referral or discussion with ophthalmology (unless via TIA/stroke pathway)
		May be painful	- Optic neuritis (e.g. MS) (may have loss of colour vision or relative afferent pupillary defect) (in practice, may not be that painful). - Giant cell arteritis (eye not painful but may have temporal pain or a headache).	
		Painless	- Mainly retinal causes: - Retinal detachment. - Vitreous haemorrhage. - Retinal vein/artery occlusion. - Wet age-related macular degeneration. - Amaurosis fugax (if transient): urgent TIA referral.	
	Bilateral	Painless	- CVA: urgent stroke referral. - Raised intracranial pressure causing papilloedema (may have red flag headaches, e.g. thunderclap headache, headaches worsened by Valsalva manoeuvre, headaches with neurological deficits or weight loss).	
Gradual onset (over months)	Unilateral or bilateral	Painless	- Refractive error correcting with pinhole. - Diabetic/hypertensive retinopathy. - Cataracts. - Open-angle glaucoma. - Dry age-related macular degeneration. - Other systemic diseases (usually inflammatory conditions). - Iatrogenic (hydroxychloroquine, amiodarone).	Usually routine ophthalmology referral (opticians for refractive errors) (obviously, manage underlying conditions such as diabetes and hypertension too!)

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